Positive Square Energy of Every Pentacyclic Graph

AI-generated manuscript; no human author is claimed

3 August 2026

Abstract

Let G be a finite simple connected graph with n vertices and $n+4$ edges. We prove that the sum $s^+(G)$ of the squares of its positive adjacency eigenvalues is at least n. Positive cyclic-block ranks have the seven partitions $1^5, 2+1^3, 2+2+1, 3+1+1, 3+2, 4+1, 5$. For the first six, an exact block-additive doubly nonnegative sieve leaves four incidence families. We give an owner-exact induced-territory closure, including positive connector routes, repeated owners, nested shared-cut attachments, and the noncanonical all-length gate. A single rank-five block suppresses to exactly 118 loopless no-cut-vertex multigraph kernels, in order counts $1, 3, 13, 24, 38, 23, 16$. Exact finite certificate ledgers, symbolic equality templates, and structural packets cover every physical parity orbit and every one-coordinate frontier. Exact path elimination and fixed-parity monotonicity promote those frontiers to arbitrary subdivisions; one-vertex additivity includes arbitrary rooted trees. We record the master tables, checksums, and the precise scope of every executable dependency.

1 Statement and analytic foundations

All graphs are finite, simple, and undirected unless a suppressed kernel is explicitly called a multigraph. If $A(G)$ has eigenvalues $\lambda_1, \ldots, \lambda_n$, set

$$s^+(G) = \sum_{\lambda_i > 0} \lambda_i^2, \qquad s^-(G) = \sum_{\lambda_i < 0} \lambda_i^2, \qquad \sigma(G) = s^+(G) - |V(G)|.$$

Since $\operatorname{tr} A^2 = 2|E(G)|$, a pentacyclic graph satisfies

$$s^+(G) + s^-(G) = 2n + 8. \tag{1}$$

Theorem 1.1. *If G is a finite simple connected graph with $|V(G)| = n$ and $|E(G)| = n+4$, then*

$$\boxed{s^+(G) \geq n}.$$

Lemma 1.2 (Induced superadditivity)**.** *If $V(G) = V_1 \,\dot\cup \cdots \dot\cup\, V_t$, then*

$$s^+(G) \geq \sum_i s^+(G[V_i]), \qquad \sigma(G) \geq \sum_i \sigma(G[V_i]).$$

Proof. For every real symmetric matrix A,

$$\operatorname{tr}(A_+^2) = \max_{X \succeq 0} \{2 \operatorname{tr}(AX) - \operatorname{tr}(X^2)\}.$$

Insert the block diagonal matrix with blocks $A[V_i]_+$. Entries of A between distinct parts do not contribute to the trace. $\square$

For a graph $H=(V,E)$ with $E\neq\varnothing$, define

$$\kappa(H)=\sup_{\substack{M\succeq 0,\ M\geq 0\\ \mathbf{1}^T M\mathbf{1}>0}}\frac{4(\sum_{uv\in E(H)}\sqrt{M_{uv}})^2}{\mathbf{1}^T M\mathbf{1}}. \tag{2}$$

Here $M\geq 0$ means entrywise nonnegative, and every matrix in this section is real symmetric with rows and columns indexed by V. Set $\kappa(H)=0$ when $E=\varnothing$.

Lemma 1.3 (DNN correlation dual)**.** *For every graph H,*

$$s^-(H)\leq\kappa(H)=\min_{\substack{R\succeq 0\\ R_{vv}=1}}\sum_{uv\in E(H)}\frac{2}{1-R_{uv}}, \tag{3}$$

where the minimum is over the compact elliptope $\mathcal{E}_V=\{R\in\mathbb{S}^V : R\succeq 0,\ R_{vv}=1\ (v\in V)\}$, and a summand is $+\infty$ when $R_{uv}=1$. If $H=H_1\vee H_2$ is a one-vertex sum, then $\kappa(H)=\kappa(H_1)+\kappa(H_2)$. Hence $\kappa(T)=|E(T)|$ for a tree.

Proof. Write $A=P-B$ for its positive and negative spectral parts. By the Schur product theorem $M=B\circ B$ is doubly nonnegative, and $\mathbf{1}^T M\mathbf{1}=\operatorname{tr}B^2=s^-(H)$. Moreover

$$s^-(H)/2=-\sum_{uv\in E(H)}B_{uv}\leq\sum_{uv\in E(H)}\sqrt{M_{uv}},$$

which gives the first inequality. If R is a correlation matrix, then

$$\mathbf{1}^T M\mathbf{1}-\sum_{uv\in E(H)}2(1-R_{uv})M_{uv}=\langle R,M\rangle+2\sum_{uv\notin E(H),\,u<v}(1-R_{uv})M_{uv}\geq 0.$$

If $R\in\mathcal{E}_V$ has $R_{uv}<1$ on E, weighted Cauchy–Schwarz and the preceding display give

$$4\Big(\sum_{e\in E}\sqrt{M_e}\Big)^2\leq\langle J,M\rangle\sum_{uv\in E}\frac{2}{1-R_{uv}}.$$

The assertion is automatic if an edge entry of R is one. Taking the two extrema proves $\kappa(H)$ is at most the right side of (3).

We prove the reverse inequality without an infinite-dimensional separation. Normalize $\langle J,M\rangle=1$ and put

$$\mathcal{D}=\{M\in\mathbb{S}^V : M\succeq 0,\ M\geq 0,\ \langle J,M\rangle=1\},\qquad \Delta_E^\circ=\{a\in\mathbb{R}^E : a_e>0,\ \sum_e a_e=1\}.$$

The set $\mathcal{D}$ is compact. The elementary identity

$$\Big(\sum_e\sqrt{x_e}\Big)^2=\inf_{a\in\Delta_E^\circ}\sum_e\frac{x_e}{a_e}\quad(x_e\geq 0)$$

and finite-dimensional convex minimax (first on $a_e\geq\varepsilon$, then $\varepsilon\downarrow 0$) yield

$$\frac{\kappa(H)}{4}=\inf_{a\in\Delta_E^\circ}\ \max_{M\in\mathcal{D}}\sum_{e\in E}\frac{M_e}{a_e}. \tag{4}$$

For fixed a, let $C(a)_{uv} = 1/(2a_{uv})$ on edges and zero elsewhere, with zero diagonal. The inner problem and its finite-dimensional SDP dual are

$$\begin{aligned} &\max \langle C(a), M\rangle && M \succeq 0,\ M \geq 0,\ \langle J, M\rangle = 1, \\ &\min \lambda && \lambda J - C(a) = S + N, \quad S \succeq 0,\ N \geq 0. \end{aligned} \tag{5}$$

There is no closure qualification here. The primal is strictly feasible with a positive multiple of $(1-\delta)I + \delta J$, where $0 < \delta < 1$; the dual is strictly feasible by taking $S = \eta I$ and then λ so large that $N = \lambda J - C(a) - S$ is entrywise positive. Slater's theorem therefore gives attainment and equality of the two values in (5).

Take a dual feasible triple. Necessarily $\lambda > 0$, and

$$R = \lambda^{-1} S + \operatorname{diag}\bigl(1 - \lambda^{-1} S_{vv} : v \in V\bigr)$$

belongs to $\mathcal{E}_V$: indeed $0 \leq S_{vv} \leq \lambda$. On an edge, $N_{uv} \geq 0$ gives $1 - R_{uv} \geq 1/(2\lambda a_{uv})$. Consequently

$$\sum_{uv \in E} \frac{2}{1 - R_{uv}} \leq 4\lambda \sum_{e \in E} a_e = 4\lambda.$$

Minimize first in (5) and then in (4). This proves the missing inequality and hence the exact dual identity. It also shows that the displayed elliptope minimum is attained whenever $E \neq \varnothing$.

For a one-vertex sum, restriction of a correlation matrix proves $\kappa(H) \geq \kappa(H_1) + \kappa(H_2)$. Conversely, choose minimizing Gram representations, identify their common unit vector, and place their components orthogonal to that vector in mutually orthogonal spaces. The glued correlation has the two prescribed principal submatrices, so the objective adds and gives the reverse inequality. Iterate $\kappa(K_2) = 1$ for the tree assertion. □

Lemma 1.4 (Exact path elimination)**.** *Replace a kernel edge uv by a path of integer length $\ell \geq 1$. For fixed endpoint correlation $r \in [-1, 1]$, its exact contribution in excess of ℓ is*

$$f_\ell(r) = \ell \tan^2\left(\frac{\arccos((-1)^\ell r)}{2\ell}\right). \tag{6}$$

For fixed r and parity, $f_{\ell+2}(r) \leq f_\ell(r)$. Thus if a subdivided kernel has total path length L,

$$\kappa(H) - L = \min_R \sum_P f_{|P|}(R_{u_P v_P}), \tag{7}$$

with R ranging over the elliptope on the branch vertices. Formula (6) is extended-valued at the endpoints (in particular, a denominator-zero edge cost is $+\infty$), and the minimum in (7) is attained.

Proof. Represent R by unit vectors and alternately negate vectors along the path. The transformed endpoint angle is $\beta = \arccos((-1)^\ell r)$. The spherical triangle inequality and convexity of $\sec^2(x/2)$ make equal successive angles optimal; equally spaced planar vectors attain the value $\ell \sec^2(\beta/(2\ell)) = \ell + f_\ell(r)$. Different paths are compatible because only endpoint vectors are shared. For $h_s(x) = s \tan^2(x/s)$, where $s > 0$ and $0 \leq x < \pi s/2$, differentiation with respect to the path-length variable s gives, for $z = x/s$,

$$\frac{\partial}{\partial s} h_s(x) = \tan z \sec^2 z (\sin z \cos z - 2z) \leq 0,$$

with strict inequality when $x > 0$. Apply this with $x = \frac{1}{2}\arccos((-1)^\ell r)$; an endpoint value is understood by a one-sided limit. This proves fixed-parity monotonicity. □

If a rank-five core has L edges, it has $L-4$ vertices. Consequently

$$\kappa(H) \leq L+4 \Longrightarrow s^{+}(H) = 2L - s^{-}(H) \geq L-4 = |V(H)|. \tag{8}$$

If a tree is joined to a DNN-certified core at one root, Lemma 1.3 adds exactly one to κ, L, and $|V|$ for each new tree edge. Thus the DNN budget is uniform under arbitrary rooted-tree attachments. This statement does not apply to a structural certificate, whose attachments are handled by induced-territory ownership instead.

We use the following packet ledger. Here $T = C_3$, $P = C_5$, and $D = \Theta(1,2,2)$; a word records cyclic blocks in one connected induced territory, not a formal sum of overlapping subgraphs. Every line permits arbitrary rooted trees, and the mixed words permit arbitrary bridge connectors. The extra "shared" hypothesis means one bridge-free shared-cut incidence tree.

packet	credit	proof owner and exact scope
rank $2,3,4$	$\sigma \geq 0$	the complete lower-rank graph theorems [1, 2, 3]
T, P	>0, $\geq 2-\sqrt{5} > -1$	rooted unicyclic cycle calculation [1, 4]
TT, TP	>1, $>3-\sqrt{5} > 0$	two-cycle cactus packets; the first also covers the attached diamond D
TTP	>0	if separated, use $TT+P$ or $T+TP$; if shared, the rooted TTC_5 phase gives $>4-\sqrt{5}$ [2, 4]
shared T^4	>3	negative Sachs phase when packing is at most two; the sole packing-three central-triangle/three-petal type is the exact matching-injection calculation of [5]
shared $D+T^k$, $0 \leq k \leq 2$	$>1+k$	grouped rooted Sachs phase; in particular $D+T>2$ and $D+TT>3$ [8]
attached K_4	>2	the standalone attached-K_4 Sachs packet [2, 3]

Table 1: Packet inequalities used in the multiblock proof. A shared cut is owned by one displayed territory only.

For clarity, the TTP line is not obtained by adding three merely strict inequalities. Its three-block reduced tree either has a separating edge, giving the displayed $TT+P$ or $T+TP$ ledger, or is one shared cluster, where the cited TTC_5 determinant has negative phase. Likewise the $D+TT$ line is the rooted activity proof: after eliminating attached trees, every one-cycle term in the normalized Sachs expansion is negative imaginary and the possible two-cycle terms are real, so $D(H) = s^{+}(H) - s^{-}(H) > 0$ and $\sigma(H) = 3 + D(H)/2 > 3$. The shared T^4 proof uses the same phase argument except for its one packing-three incidence, treated by the cited injection. These are the exact strengths used below; no uniform margin is inferred from strict positivity. A nonempty tree has credit -1 because its spectrum is symmetric and $s^{+}(T) = |E(T)|$.

2 Exhaustive block partitions and the exact multiblock sieve

Cyclomatic rank is additive over cyclic blocks. Its positive block ranks therefore have exactly the partitions

$$1^5, \quad 2+1^3, \quad 2+2+1, \quad 3+1+1, \quad 3+2, \quad 4+1, \quad 5. \tag{9}$$

A rank-one block is a cycle. Suppression in a rank-two block and $\sum_v (d(v)-2) = 2$ show that it is a theta: three parallel kernel paths.

For a cycle define

$$\epsilon_q = \begin{cases} 0, & q \text{ even}, \\ q\tan^2(\pi/(2q)), & q \text{ odd}, \end{cases}$$

so $\epsilon_3 = 1$, $\epsilon_5 = 5 - 2\sqrt{5} < 2/3$, and $\epsilon_q < 2/5$ for odd $q \geq 7$. If a theta has path lengths a, b, c, exact path elimination gives an excess $\Delta(a, b, c)$. The elementary tangent classification is

$$\Delta > 1 \iff (a, b, c) = (1, 2, r),\ r \geq 2; \quad \max \Delta = \Delta(1, 2, 2) = \frac{\sqrt{17} - 1}{2}, \tag{10}$$

and every other exceptional row has $\Delta < 4/3$. This follows by replacing each nonunit odd path by length three and each even path by length two using Lemma 1.4; the one-unit objective has $\cot^2 x + 2\tan^2(x/2) \geq 1$, with equality analysis leaving precisely $(1, 2, r)$. Minimizing the $r = 2$ expression $1/(4u^2) - 1/2 + 17u^2/4$ gives the displayed maximum. These details also show that physical path lengths, rather than switching classes, are being used.

Proposition 2.1 (Exact multiblock sieve)**.** *After block additivity of κ and all available exact lower-rank ledgers, the rank partitions with at least two cyclic blocks have precisely the residual rows in Table 2; every omitted row has total DNN excess at most four.*

Proof. For $2+1^3$, order the cycle excesses. If the third is below one, the bounds $\epsilon_5 < 2/3$ and $\epsilon_{q\geq 7} < 2/5$, together with $\Delta < 4/3$ away from $\Theta(1, 2, r)$, give the direct inequality. In the exceptional theta family, three triangles survive; replacing one by a nontriangle survives only at the unique maximum D and only for P. For $2+2+1$, a nonmaximum theta has excess $< 4/3$, while $2(\sqrt{17}-1)/2+\epsilon_5 < 4$; hence exactly $D + D + T$ remains.

For rank three, the cited physical ledger has $e_B \leq 2$ in every direct row. In class 111, lengthening a doubled member gives respectively $229/120, 31/20, 1862/1000, 1662/1000 < 2$; the only canonical structural rows are the two doubled families, one-long all-odd K_4, and the actual K_4. Consequently two cycles can survive only as stated. Against a theta, every structural row except the actual K_4 has $e_B < 12/5$, and $12/5 + (\sqrt{17} - 1)/2 < 4$. Finally the rank-four theorem partitions every row into a direct certificate with $e_B \leq 3$ or its displayed structural list, so $\epsilon_q \leq 1$ proves the direct part. The exact five-cycle calculation and the cactus census own 1^5. Thus every physical row has exactly one ledger owner; switching is never used to change its canonical lengths, and no edge-addition monotonicity is used. □

Proposition 2.2 (Multiblock closure)**.** *Every connected pentacyclic graph with at least two positive-rank cyclic blocks satisfies $s^+(G) \geq |V(G)|$.*

Proof. The cycle-only row is the complete pentacyclic cactus theorem [5]. The remaining routine rows have the following ownership ledger. "Boundary open" means that the cut vertex stays with the anchor and the cycle minus that vertex, including every branch rooted away from the cut, is the other induced territory. At an actual bridge the whole descendant suffix of the rooted block-cut tree, including the bridge-side endpoint and all connector vertices, stays together.

row	territory construction	credit ledger
$\Theta(1,2,r)+T^3$	open an internal vertex of the long arm, with its complete owned descendants; both arm remnants stay with the endpoint side	if it owns a triangle, opened side >0 and complement ≥ 0; otherwise tree -1, while the retained T^4 side is >3 when shared, or bridge-splits as $TT+T+T$ with credit $>1+0+0$
$D+D+T$	across a bridge use complete descendant sides; at a shared cut use the two cut-vertex orbits of the standard diamond opening, keeping the cut and one attached diamond on the anchor side	the second side is a tree -1 unless it owns T, when it is >0; otherwise the anchor is shared $D+T$ and is >2
$K_4+Q_1+Q_2$	keep the complete attached K_4 and root the reduced block tree there; each boundary component is kept whole, or its first cycle is boundary opened; a component containing both demands is never split at both ends	anchor >2; each of at most two components has credit ≥ -1, while a rank-two component has ≥ 0
$K_4+\Theta$	keep the complete attached K_4; at a shared cut z, keep z with it and put $\Theta-z$ and all theta-side branches together	anchor >2; the other side is a tree or unicyclic territory, hence ≥ -1
structural $4+1$	use the physical rank-four opening: its chosen vertex and descendants form one tree, while path remnants stay with the favorable rank-three anchor; keep the external cycle intact or boundary open it	anchor >2 pays the opening tree and at most one further -1 boundary territory

In each line the described owner classes are disjoint and exhaustive; every rooted branch follows its root. Table 1 proves every displayed credit, including the packing-four subcase in the first line. This closes all routine residuals without a formal sum over overlapping packets.

It remains to repair, owner-exactly, the four rows

$$D+C_3+C_3+C_5,\quad B_{\mathrm{DT}}+C_3+C_3,\quad B_{\mathrm{DC4}}+C_3+C_3,\quad B_{K_4}+C_3+C_3. \tag{11}$$

We now define the physical openings and owners used in the four repairs. Deleting the edges of B, every off-B component meets B in at most one vertex; otherwise it and a path in the 2-connected block B would form a larger cyclic block. Its unique meeting vertex, or the first vertex of B on its block-cut route, is its *owner*. An opening is physical: it deletes one actual internal path vertex and all descendants owned there; it does not switch a parity row. The canonical opening and legal first owners are exactly:

B	admissible physical opening	legal first owners after no routed opening
D, paths xy, xay, xby	a or b	the endpoints x, y
doubled triangle 111, canonical pairs, connector length ≥ 3	an internal vertex of the odd connector	all branch vertices and all internal vertices outside that connector
doubled triangle 111, direct connector	the internal vertex of either canonical even member	the three branch vertices
doubled-C_4 111, canonical pairs	an internal vertex of every available single connector (in particular the normalized even connector)	the four branch vertices and interiors of the two canonical even doubled members
one-long all-odd K_4	an internal vertex of the unique long path	the four branch vertices

The doubled-C_4 interior owners are therefore a genuine orbit, not branch owners in disguise. For direct/direct incidence the owner datum is an ordered pair, with repetition, from the appropriate legal set. For direct/nested incidence the upstream triangle has such an owner and the downstream owner is an arbitrary physical vertex of that triangle.

Call a route *positive* precisely when the minimal block-cut subtree from B to the external cyclic blocks contains an actual bridge. Root that subtree at B and choose a farthest bridge. Its entire descendant suffix is one connected induced territory; the complement is connected and lower-rank. The exact positive-route ledger is

family	descendant profile	credit against complement
each rank-three $B + TT$	T or TT	> 0 or > 1, while the rank-four or rank-three complement is ≥ 0
$D + TT + P$, triangle below bridge	T, TT, TP, or TTP	respectively $> 0, > 1, > 0, > 0$; the complement contains D and has rank at most four, hence ≥ 0
$D + TT + P$, only P below bridge	P	> -1, while the retained shared $D + TT$ packet is > 3

The TP and TTP strengths and their separating/shared proofs are exactly the ones in Table 1. This ledger includes separate arms, a common positive stem, and nested blocks: the suffix, never a hand-picked cycle list, owns every connector and shared cut once.

The bridge-free subtree for a rank-three block with two triangles has only two block-level forms: direct/direct, allowing repeated owners, and direct/nested. In the nested form the downstream owner is a vertex of the upstream triangle. Open the second triangle T_2 at its boundary cut, retaining the cut upstream. The remainder of T_2, with all branches rooted away from the cut, is one induced nonempty tree S, so $\sigma(S) = -1$.

For the three rank-three rows, first apply the all-length gate. In doubled triangle class 111, lengthening a member of either doubled pair has exact excess bounds $229/120 < 2$ or $31/20 < 2$; in doubled-C_4 class 111 the bounds are $1862/1000 < 2$ or $1662/1000 < 2$. Relabelling and fixed-parity monotonicity cover every noncanonical member. Adding two triangle excesses gives at most four, so these rows are DNN-closed for every owner. Hence both doubled pairs may now be assumed canonical.

Perform the canonical physical opening of B from the table. Its owned territory R is one tree. The other external triangle T_1 remains in the retained anchor A. For the doubled triangle, opening a long odd connector retains two intrinsic triangles; when that connector is direct, opening a canonical even parallel path retains a diamond. For doubled C_4, branch owners give the three-triangle anchor, while an owner internal to a canonical even doubled path gives the distinct $D+T$ anchor. For one-long all-odd K_4, opening the long path retains a diamond. In every owner orbit, adjoining T_1 therefore gives either a three-triangle favorable anchor of packing number at most two or an attached $D+T$ anchor. In either case $\sigma(A) > 2$. The partition is

$$V(G) = V(A) \,\dot\cup\, V(R) \,\dot\cup\, V(S), \qquad \sigma(G) > 2 - 1 - 1 = 0.$$

For equal direct owners the common cut stays only in A; for nested incidence all of T_1, including the downstream cut, stays in A. Structural path remnants also stay in A, and each rooted branch follows its root. Thus all three territories are connected, induced, disjoint, and exhaustive.

Finally consider the bridge-free diamond row. Its rooted classification is not merely direct/direct versus direct/nested at D. There are exactly two possibilities. First, some triangle is the first block after D; retain that $D+T$ shared-cut anchor, of credit > 2, and boundary-open the at most two remaining cyclic demands. A component containing both demands is kept as one rank-two territory or is opened once, so the total loss is at most two. Second, no triangle is first after D; then P is the first external block, and relative to P the triangles are direct/direct (owners may repeat) or direct/nested. Let u be the D–P cut, v the owner of an upstream triangle, and in the direct/direct case let w be the second owner. Choose $z \in V(P) \setminus \{u, v, w\}$, possible because at most three of its five vertices are forbidden. Put z and all descendants owned there in one tree R; the induced path $P - z$ stays in an attached $D+T$ anchor A of credit > 2. Boundary-open the other triangle into a tree S. Repeated owners cause no duplication: each boundary cut remains only in A. Hence $\sigma(G) > 2 - 1 - 1 = 0$.

In every structural split, ownership is fixed before credit is invoked. The opened vertex and complete descendants form R; the boundary cut of T_2 stays in A, while T_2 minus that cut and descendants rooted there form S; every remaining core vertex, path remnant, first triangle, and descendant belongs to A. These are connected induced, disjoint, exhaustive territories. Thus every connector, shared cut, physical-path remnant, and rooted branch has exactly one owner. The full terminal audit is recorded in [6, 7]. □

3 Classification of a single rank-five block

Suppress each maximal path whose internal vertices have block degree two, retaining parallel kernel edges. Suppression creates no loop: such a loop would be a cycle meeting the remainder at one branch vertex, making that vertex a cut of the original block. Simplicity says that at most one path in any parallel class has length one.

Proposition 3.1 (The 118 kernels)**.** *A loopless multigraph K with at least two vertices, no cut vertex, minimum degree at least three, and cyclomatic rank five has order between two and eight and is isomorphic to exactly one of 118 canonical kernels. Their order and degree ledgers are Tables 3 and 4.*

Proof. Writing $v = |V(K)|$ and $e = v + 4$ gives

$$\sum_x (d(x) - 2) = 2e - 2v = 8.$$

Every summand is positive, so $2 \le v \le 8$. For each positive partition of eight, add two to obtain a candidate degree multiset. Introduce nonnegative integer multiplicities a_{ij} and solve

$$\sum_{j \neq i} a_{\min(i,j),\max(i,j)} = d_i.$$

Reject a solution whose positive support disconnects after deletion of a vertex, and quotient by simultaneous row and column permutations. This gives exactly Table 4, and hence Table 3. Taking the lexicographically least upper-triangle multiplicity vector over every labelling proves pairwise distinction and gives the fixture `research/fixtures/rank-five-kernels.json`, whose SHA-256 is

`027c84d6dd777a29b3dc93389ab30b5d43f6507eddceb4ea286f1240da95b884.`

The independent generator reconstructs the incidence solutions before reading the fixture and requires exact equality [9]. □

4 Single-block master tables and all-length closure

A physical row records the number of odd paths in each parallel bundle; it is not a switching class. For each residual parity orbit, its finite frontier is the canonical shortest simple length vector and every vector obtained by adding two to one path. If a general vector differs in some coordinate, choose that coordinate frontier and use Lemma 1.4 in all coordinates. If it does not differ, use the canonical target. This is the frontier principle used in every line below.

Proposition 4.1 (Exact 118-kernel master theorem)**.** *Every simple subdivision B of every kernel in Proposition 3.1, with arbitrary rooted trees attached at arbitrary vertices, satisfies $s^+(G) \ge |V(G)|$.*

Proof. Tables 5 and 6 are the exhaustive proof ledger. "Tetra" denotes the exact regular-tetrahedron three-color sieve. Rational records give explicit stereographic unit vectors; the verifier reconstructs every branch and internal vector over $\mathbb{Q}$, checks each physical path, and sums exact step costs $(1-r)/(1+r) < 4$. Equality records check every principal minor of the displayed rational or quadratic Gram matrix and every path cost. Structural records invoke Lemma 1.2, never a DNN claim.

For completeness, the analytic low-order line is as follows. The order-two kernel is six parallel paths; endpoint tests after parity reduction, followed by the sole interior one-unit minimization, give strict excess below four. The three order-three multiplicities are $(1,2,4),(1,3,3),(2,2,3)$; all 98 physical rows collapse to 74 genuine automorphism orbits and carry exact rational correlations. The 13 order-four kernels have eight paths, so the 13 coarse residuals give $13(1+8) = 117$ targets.

At order five, the separated all-odd $K_5 - e$ case is essential: its optimized DNN excess is $2\sqrt{7} - 1 > 4$, so no universal DNN budget is asserted. Its 19,683-state disjunction in Table 6 ends with 16 exact residual orbits. In the all-unit structural state an actual attached K_4 is retained; it is not replaced by an arbitrary all-odd K_4 subdivision. In K22, only coordinates whose growth remains on the deleted tree side use the structural frontier; every other one-coordinate frontier is rational. The same coordinate-safe rule applies to K71 at order six.

The K80 and K118 equality records use a signed four-cycle Gram matrix: adjacent entries have absolute value $1/2$ and opposite entries are zero. The matrix is positive semidefinite (balanced signs give eigenvalues $0,1,1,2$; frustrated signs give $1 \pm 1/\sqrt{2}$, each twice). Four doubled support edges each cost $1/3+2/3 = 1$, while the listed singleton coordinates have transformed endpoint correlation one and cost zero. This proves exact total four and explains why only the listed coordinate frontiers inherit equality.

The independent implication master digest-locks the 118-row census and all six theorem-owner packages, invokes each verifier, checks its acceptance line, and requires the exact scope partition

$$(2\text{–}4), \quad (5 \text{ ordinary}), \quad (5 \text{ all-odd } K_5 - e), \quad 6, \quad 7, \quad 8.$$

Thus every kernel has an owner and the complementary order-five scopes are not silently widened. Every DNN target gives (8); every structural target directly gives the same spectral conclusion by induced superadditivity. The frontier principle gives every simple subdivision.

It remains to be explicit about arbitrary trees. There are two mechanisms, and they must not be conflated. For a DNN target, the subdivided block and each rooted tree form iterated one-vertex sums. Lemma 1.3 therefore adds exactly the number of new tree edges to κ; since the same number is added to L and to $|V|$, (8) persists. For a structural target no DNN bound is asserted. Instead its certificate specifies an induced vertex partition and an owner for every core path vertex. Assign each component of the forest off the block wholesale to the unique territory containing its attachment vertex. If the structural deletion itself lies in an attached tree, take the complete descendant subtree on the deleted side. These rules preserve disjointness, inducedness and exhaustiveness, and the packet lemmas in Table 1 already allow arbitrary rooted trees. Induced superadditivity then proves the structural descendant. This proves the proposition [10]. □

Proof of Theorem 1.1. Apply the exhaustive partition (9). If at least two cyclic blocks occur, Proposition 2.2 applies. Otherwise suppress the unique rank-five block. Proposition 3.1 gives one of the 118 kernels, and Proposition 4.1 covers its subdivision and every attached tree. These alternatives are exhaustive. □

A The finite verification as a computer-assisted proof

This appendix specifies the finite part of Proposition 3.1 and Proposition 4.1 as a computer-assisted proof. The committed JSON files named below are proof objects, not illustrative data: together with their verifiers they are machine-readable theorem appendices. Printing their more than 10^5 records would add no logical content. Completeness is instead established by regeneration or exact key-set equality, and byte identity is fixed by SHA-256.

The logical model is the standard one for a computer-assisted finite proof. The mathematical text proves that acceptance of the stated finite predicates implies the theorem; the listed deterministic programs prove those predicates by exhaustive integer/rational computation. The trust base is the displayed reduction, the verifier source, Python's specified exact integer semantics, and the hardware/interpreter executing it. SHA-256 is used only to identify the audited byte strings, never as evidence that a mathematical predicate is true. No randomness, numerical optimizer, floating-point tolerance, network service, or unrecorded search output is part of the proof. A reader may inspect and run the programs or independently reimplement the finite predicates from the pseudocode and schemas below; the certificate files are witnesses, while fresh regeneration and exact key-set comparison establish exhaustion.

A.1 Algorithms and completeness

The kernel classifier uses the following finite algorithm. Here $a_{ij} \in \mathbb{Z}_{\geq 0}$ is the upper-triangle multiplicity code.

```
for v = 2,...,8:
  for each nonincreasing positive partition (x_1,...,x_v) of 8:
    d_i = x_i + 2
    recursively enumerate every nonnegative solution of
        sum_{j != i} a_ij = d_i  (i=1,...,v)
    retain exactly those supports connected after deletion of every vertex
    replace each retained code by the lexicographically least relabelling
sort, deduplicate, and compare byte-for-byte with rank-five-kernels.json
```

The identity $\sum_i (d_i - 2) = 8$ proves that this generates every possible minimum-degree-three suppressed rank-five kernel; the vertex-deletion test is exactly the no-cut-vertex predicate. For $v = 8$ the implementation uses an equivalent recursive cubic multigraph generator, fixes the first multiplicity row, and canonicalizes over all possible root images and neighbor orders.

For a fixed kernel with bundle multiplicities m_e, a *physical row* is $o = (o_e)$ with $0 \le o_e \le m_e$, where o_e is the number of odd paths in bundle e. The order-three through order-eight censuses implement the following exhaustive scheme.

```
for each kernel K in the locked 118-row fixture:
  generate Cartesian product 0 <= o_e <= m_e
  quotient only by Aut(K), using the least transported row as key
  test every coarse exact certificate; retain every failure as a residual
for each residual key (K,o):
  generate (K,o,none) and (K,o,j) for every physical path coordinate j
require generated target keys == certificate keys (no omissions or extras)
```

Thus "physical row" does not mean a switching class. Orbit sizes are summed back to the unquotiented Cartesian-product totals. The canonical length list for a bundle is one path of length 1 and the other odd paths of length 3, with every even path of length 2; target j adds two to coordinate j. These are all the finite cases needed: if an arbitrary same-parity length vector is canonical, its canonical target applies; otherwise choose any increased coordinate j, start from target j, and decrease all remaining coordinates by Lemma 1.4. This is a proof of the infinite all-length reduction, not a graph-order cutoff.

The implication master then checks the disjoint owner registry

$$\{2, 3, 4\} \,\dot\cup\, (\{5\} \setminus \{K_5 - e \text{ all odd}\}) \,\dot\cup\, \{K_5 - e \text{ all odd}\} \,\dot\cup\, \{6\} \,\dot\cup\, \{7\} \,\dot\cup\, \{8\}.$$

It digest-locks and executes every owner, checks a required acceptance line, and records a canonical JSON dependency manifest. Equality of the generated and certified key sets proves coverage; distinct canonical keys and the disjoint registry prove nonduplication. Table 5 gives the resulting supplemental count ledger, including all 108467 order-five through order-eight frontier targets, so the individual rows need not be typeset.

A.2 Certificate formats and exact predicates

All JSON integers are checked to be genuine integers; a rational is stored as a reduced pair $[p, q]$ with $q > 0$ (the small $K_5 - e$ appendix uses an equivalent canonical rational string). Four certificate types occur.

1. A tetrahedral record stores a physical key, orbit size, coloring, and a rational upper bound. The verifier recomputes the automorphism orbit and the bound in $\mathbb{Q}$ and accepts a coarse row exactly when the bound is at most 4.

2. A strict rational path-vector record stores stereographic parameters $t \in \mathbb{Q}^d$. The verifier reconstructs
$$u(t) = \frac{(1 - \|t\|^2, 2t)}{1 + \|t\|^2}, \qquad c(u, v) = \frac{1 - u \cdot v}{1 + u \cdot v},$$
checks every branch and internal vector, path width and parity-transformed endpoint, recomputes the sum of all step costs in $\mathbb{Q}$, requires equality with the stored reduced rational, and finally requires the strict integer predicate numerator$(C) < 4\,$denominator(C). No floating-point comparison is used.

3. A symbolic equality record stores a rational (or explicitly quadratic) Gram matrix. The verifier checks symmetry, unit diagonal, and every principal minor nonnegative by fraction-preserving elimination, then recomputes each path cost and the identity $C = 4$. The K80 and K118 signed-cycle records additionally reconstruct their sign transport from the parity row.

4. A structural record stores deleted and retained branch/path indices. The verifier reconstructs the physical simple subdivision, checks incidence, connectivity, disjoint induced territories and the claimed retained K_4 or triangle packet, and then invokes the stated mathematical packet lemma. Such a record is never accepted as a DNN certificate.

For every source file the verifier first checks the schema, canonical encoding, scope and SHA-256. It rejects duplicate, absent, extra, malformed or out-of-range keys; verifies the displayed count partition; and compares a frozen theorem fixture with a fresh deterministic reconstruction wherever that fixture is derived from smaller sources. Order eight streams four disjoint chunks and uses a bit vector indexed by residual and frontier to prove that every expected key occurs exactly once.

A.3 Frozen theorem appendices

The principal committed proof objects and their SHA-256 values are listed in Table 7. The larger rational-result and order-eight chunk hashes are source-locked in the corresponding verifier and are also reported below. These paths, schemas and hashes constitute the unabridged appendices.

A.4 Hostile checks and reproducibility

Every top-level verifier uses an explicit `require` that raises an exception, rather than relying on Python's removable `assert` statement. The suites deliberately mutate deleted and duplicated records, multiplicities, ordering, counts, parity and frontier keys, rational costs, Gram entries, structural ownership, source hashes, and theorem scopes, and require every mutation to be rejected. The classification, low-order and implication masters reject respectively 9, 8, and 9 attacks; the lower order-three and order-four suites reject 11, 9, and 10; the order-five through order-eight owner suites report their own attack totals. The masters also run under `python3 -O` and require byte-identical normal and optimized output, so optimization cannot remove a proof predicate.

Run the classification and single-block implication master from the repository root with Python 3.10 or newer, using only the standard library, in normal and optimized modes:

```
python3 research/rank-five-kernel-census-verifier.py
python3 -O research/rank-five-kernel-census-verifier.py
python3 research/rank-five-order2-8-master-verifier.py
python3 -O research/rank-five-order2-8-master-verifier.py
```

Each implication-master command recursively invokes the low-order verifier and the order-five through order-eight theorem verifiers, including the separate all-odd $K_5 - e$ owner. To audit an individual package directly, run:

```
python3 research/rank-five-low-order-master-verifier.py
python3 pentacyclic/research/order5-kernel-family-theorem-verifier.py
python3 pentacyclic/research/all-odd-k5e-theorem-verifier.py
python3 pentacyclic/research/order6-kernel-family-theorem-verifier.py
python3 pentacyclic/research/order7-kernel-family-theorem-verifier.py
python3 pentacyclic/research/order8-kernel-family-theorem-verifier.py
```

Repeat each command with `python3 -O`. A successful implication-master run prints the kernel count $1+3+13+24+38+23+16 = 118$, the exact owner scopes, a dependency-manifest SHA-256, and the number of rejected hostile mutations. Order-eight rational chunks additionally have hashes

```
2b4f2ccdb91c4cb6e8f27da94f99afec24d36c0fd0fc0175683ee9e97cd901f3
4714d80c3e3b5161add1f915378af0a9541181b210bafd64ce531678c6bb0929
291beab34f3d3d3649042eea2ce55c76657aa460d59ace71615fa133b6a2b81a
09e0d6f38b19177b54f7579d14bc05dfc37bfb6f1c5308d870106698130feccf.
```

The programs prove exact finite statements: incidence enumeration, automorphism-orbit and key exhaustion, rational arithmetic, Gram principal minors, symbolic path costs, source hashes, and owner registries. They do not replace Lemmas 1.2–1.4, the all-length monotonicity argument, the block partition, or the multiblock owner proof. Conversely, the paper does not infer an all-length theorem from a finite graph-order census or a floating optimizer. The multiblock proof is mathematical and its full owner-by-owner source is [6]; it has no claimed executable verifier.

Build with

```
bash scripts/build-paper.sh all-pentacyclic-graphs
```

Scoped nonclaims. The theorem concerns finite simple connected graphs with exactly $m = n + 4$. It does not assert edge monotonicity of s^+, a universal DNN excess-four bound, strict inequality, or a classification of equality. In particular, the all-odd $K_5 - e$ family is deliberately structural/disjunctive, and several auxiliary DNN certificates attain excess four.

AI disclosure

This manuscript and its proof-object synthesis were generated by an AI coding and mathematical reasoning system from the cited repository materials. No human authorship, review, publication, or independent verification is claimed. The result should be assessed from the displayed argument and exact local dependencies.

References

[1] Companion manuscript, *Positive Square Energy of Every Bicyclic Graph*, 2026; `all-bicyclic-graphs/paper.tex`.

[2] Companion manuscript, *Positive Square Energy of Every Tricyclic Graph*, 2026; `all-tricyclic-graphs/paper.tex`.

[3] Companion manuscript, *Positive Square Energy of Every Tetracyclic Graph*, 2026; `all-tetracyclic-graphs/paper.tex`.

[4] Companion manuscript, *Positive Square Energy of Every Tetracyclic Cactus*, 2026; `all-tetracyclic-cacti/paper.tex`.

[5] Companion manuscript, *Positive Square Energy of Every Pentacyclic Cactus*, 2026; `all-pentacyclic-cacti/paper.tex`.

[6] *Four multiblock residuals: owner-exact closure*, proof note, 2026; `positive-square-energy/pentacyclic-general/four-residual-owner-exact-closure.md`.

[7] *Four residual templates: exact block-cut terminal reduction*, proof note, 2026; `positive-square-energy/pentacyclic-general/four-residual-block-cut-terminal-reduction.md`.

[8] *Canonical doubled-C_4 class* 111 *plus two triangles*, proof note, 2026; `positive-square-energy/pentacyclic-general/doubled-c4-111-two-triangles-retained-packet.md`.

[9] *Classification of the rank-five suppressed kernels*, proof note and exact fixture, 2026; `positive-square-energy/pentacyclic-general/rank-five-kernel-classification.md`.

[10] *Master theorem for all 118 rank-five suppressed kernels*, proof note and dependency audit, 2026; `positive-square-energy/pentacyclic-general/rank-five-order2-8-master-theorem.md`.

partition	direct DNN test	exact residual before territory repair	provenance of the exhaustion
1^5	$\sum \epsilon_q \leq 4$	T^4Q and T^3P^2	exact cycle ledger and complete pentacyclic-cactus census [5]
$2+1^3$	$\Delta + \sum_{i=1}^{3} \epsilon_{q_i} \leq 4$	$\Theta(1,2,r)+T^3$; $D+T+T+P$	theta classification (10) and $\epsilon_3, \epsilon_5, \epsilon_{q\geq 7}$
$2+2+1$	$\Delta_1 + \Delta_2 + \epsilon_q \leq 4$	$D+D+T$	the unique theta maximum in (10)
$3+1+1$	$e_B + \epsilon_p + \epsilon_q \leq 4$	canonical doubled-triangle 111, canonical doubled-C_4 111, or one-long all-odd K_4, each with TT; also $K_4 + Q_1 + Q_2$ if $\epsilon_1 + \epsilon_2 > 1$	the complete physical rank-three ledger, including the four noncanonical rational gates [3, 6]
$3+2$	$e_B + \Delta \leq 4$	unsubdivided $K_4 + \Theta(1,2,r)$	all other structural rank-three rows have $e_B < 12/5$
$4+1$	$e_B + \epsilon_q \leq 4$	the rank-four structural rows plus one cycle	the complete rank-four direct/structural ledger [3]

Table 2: Complete excess-four multiblock residual sieve.

$\lvert V(K)\rvert$	2	3	4	5	6	7	8	total
kernels	1	3	13	24	38	23	16	118

Table 3: Rank-five canonical kernel counts.

degrees	count	degrees	count	degrees	count
$6,6$	1	$6,5,3$	1	$6,4,4$	1
$5,5,4$	1	$6,4,3,3$	2	$5,5,3,3$	4
$5,4,4,3$	4	$4,4,4,4$	3	$6,3,3,3,3$	2
$5,4,3,3,3$	11	$4,4,4,3,3$	11	$5,3,3,3,3,3$	7
$4,4,3,3,3,3$	31	$4,3,3,3,3,3,3$	23	$3,3,3,3,3,3,3,3$	16

Table 4: Degree-multiset refinement; counts sum to 118.

order	kernels	physical rows	orbits	coarse direct	exact residual closure
2	1	–	–	analytic	six-path tangent theorem, excess < 4
3	3	98	74	–	74 exact rational orbit records, maximum ≤ 4
4	13	1,281	821	808	13 orbits; 117 targets = 116 strict rational +1 symbolic equality
5	24	6,282	4,238	4,030	207 ordinary residual orbits give 2,070 targets: 2,062 strict, 4 equality, 4 structural; all-odd $K_5 - e$ separated
6	38	23,208	12,810	11,312	1,498 orbits, 16,478 targets: 16,451 strict, 18 equality, 9 structural
7	23	31,112	18,026	14,306	3,720 orbits, 44,640 targets: 44,616 strict, 24 equality
8	16	46,736	11,188	7,705	3,483 orbits, 45,279 targets: 45,249 strict, 30 equality

Table 5: Master exhaustion by kernel order. Dashes denote a differently parameterized analytic or exact orbit ledger, not an omitted case.

scope	exceptional exact objects	closure and descendant rule
orders 2–4	six-dipole; 13 four-vertex residuals	six-path convexity; 117-element frontier, with one exact equality
order 5 ordinary	K35 equalities; K22 structural rows	four equality targets; four actual-K_4-against-tree targets; all other target coordinates strict rational
all-odd $K_5 - e$	19,683 unit/residue states	$18,848$ simplex $+53$ actual K_4 $+640$ theta $+142$ residual; the latter form 16 orbits with strict rational records
order 6	K55, K61, K71	9 equality targets each for K55 and K61; 9 K71 triangle-plus-actual-K_4 structural targets; other coordinates strict
order 7	K80 signed-cycle support	six parity rows at canonical and coordinates $0, 3, 6$: 24 equality keys; remaining coordinates strict
order 8	K118 signed-cycle support	six parity rows at canonical and coordinates $0, 5, 6, 11$: 30 equality keys; remaining coordinates strict

Table 6: Exceptional master objects and exact composition.

machine-readable theorem appendix and SHA-256
`research/fixtures/rank-five-kernels.json`
`027c84d6dd777a29b3dc93389ab30b5d43f6507eddceb4ea286f1240da95b884`
`research/fixtures/rank-five-three-vertex-orbits.json`
`e3ec57422ba2d9ca0c25ad2ba7d85b8bc74a5d656ebfe20fdb072a0688d01fa9`
`research/fixtures/rank-five-four-vertex-tetrahedral-sieve.json`
`0b8ded3f4dbe0b8de916c085393c5f470bbaf8961deddf4305396e15f1d45588`
`research/fixtures/rank-five-four-vertex-residual-frontiers.json`
`09a7b38b1e9f5e18aaddc1f9e0114b8490151f2062d3f51100c52eb314eb56d2`
`pentacyclic/research/all-odd-k5e-theorem.json`
`35523cc3be872181e2f343a7e21936f82b14e4a6968896fc2dcfd5f545da1ee1`
`pentacyclic/research/order5-kernel-family-theorem.json`
`4d8b826b397dc269c7853b8bd386d00bf469282b52720b8dac96d850e9e616d8`
`pentacyclic/research/order6-kernel-family-theorem.json`
`69b236b014aef58c037c610ca01fa62ad82601f7bb34153939ec4ddd3b5f364d`
`pentacyclic/research/order7-kernel-family-theorem.json`
`1de37116d406f72abba33f85678be9f2eba38e71347a79c67bad5f159e2f1c16`
`pentacyclic/research/order8-kernel-family-theorem.json`
`f1c08641de224194d871197454d7056eb0884c8972c535dd8daa5abd08a37a6f`

Table 7: Committed machine-readable theorem appendices.